

Satnam Singh

Full Stack Web Developer

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Moga, Punjab, India

PROFESSIONAL SUMMARY

Full Stack Web Developer with 1+ year of hands-on experience building AI-powered web applications and scalable digital products. Proficient in integrating modern AI APIs, including Google Gemini, OpenAI GPT, xAI Grok, and Anthropic Claude. Skilled in developing full-stack solutions using Next.js, React.js, Node.js, TypeScript, and modern databases. Strong focus on writing clean, maintainable code and collaborating within agile teams to deliver high-quality, user-centric applications.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python
Frontend:	React.js, Next.js 14, Tailwind CSS, HTML5, CSS3
Backend:	Node.js, Express.js, REST APIs, tRPC, JWT Authentication
Databases:	MongoDB, MySQL, Prisma ORM
AI / APIs:	Google Gemini, OpenAI GPT, xAI Grok, Anthropic Claude, Stripe
DevOps & Tools	Git, GitHub, Vercel, Vercel Cron Jobs, Puppeteer, VS Code
Practices	Agile/Scrum, CI/CD, Role-Based Access Control, Type Safety, Code Review

PROFESSIONAL EXPERIENCE

Full Stack Developer

May 2025 – Present

[GK Next Solutions](#) | Moga, Punjab, India

- Design, develop, and maintain full-stack web applications using Next.js, Node.js, TypeScript, and Tailwind CSS, delivering secure, scalable, and responsive solutions for production environments.
- Integrated multiple AI APIs (Google Gemini, OpenAI GPT, xAI Grok, Anthropic Claude) to engineer intelligent, data-driven application features, including dynamic report generation pipelines and AI-powered chat interfaces.
- Architected and implemented JWT-based authentication systems with role-based access control (RBAC), securing user data and enforcing granular access policies across all application layers.
- Built RESTful and tRPC APIs connecting frontend interfaces to MongoDB and MySQL databases, optimizing query performance and ensuring type-safe end-to-end data flow.
- Collaborated with cross-functional teams throughout the full software development lifecycle — from requirements gathering and system design to deployment and post-launch debugging.
- Managed background job processing using Vercel Cron Jobs and custom script timeouts to handle long-running AI tasks asynchronously, ensuring seamless user experience under heavy workloads.

PROJECTS

The Watch-AI — AI Research Report Platform · [Next.js 14](#), [TypeScript](#), [tRPC](#), [Tailwind CSS](#), [MySQL](#)

- full-stack AI-powered research platform that executes configurable multi-step pipelines to generate deep, structured research reports across six specialized analysis modes: Current Developments, Foundational Analysis, Anticipatory Analysis, Actionable Analysis, Argumentative Analysis, and Data Mining.
- Engineered a flexible AI orchestration layer enabling users to configure each pipeline step independently — specifying the AI model (Gemini, GPT, Grok, Claude), system prompt, max tokens, temperature, web search, and file attachments — for precise, context-aware output.
- Delivered 8+ export formats per report (PDF, DOCX, HTML, TXT, CSV, Mind Map, Chart, PowerPoint), maximizing accessibility and downstream usability of generated insights.
- Implemented secure JWT authentication with RBAC, background cron job processing, AI response history tracking, token usage management, and file upload/output handling for a production-ready, enterprise-grade platform.

Effective Agents — AI Agent Collaboration Platform · [Next.js](#), [TypeScript](#), [Tailwind CSS](#)

- Collaborated with a development team to design and ship new product features, code reviews, and feature delivery.
- Identified and remediated security vulnerabilities and critical bugs across the codebase, improving overall application stability and reducing production incident rates.
- Performed end-to-end feature testing, documenting defects and verifying fixes to uphold release quality standards.

Camson Crown — Data Ingestion & Automation Tool · [Next.js](#), [TypeScript](#), [Tailwind CSS](#), [Puppeteer](#)

- Built automated data ingestion pipelines to parse, validate, and persist user and municipality data sourced from structured email inputs into a relational database, eliminating manual data-entry overhead.
- Developed a Puppeteer-based screenshot automation utility integrated with a cron scheduler, enabling scheduled visual capture of web pages for auditing and reporting workflows.
- Designed and implemented reusable React frontend components and Next.js REST API routes to support seamless third-party API integrations.

EDUCATION

Bachelor of Computer Applications (BCA)

2021 – 2024

[Baba Mangal Singh College](#) | MRSPTU, Punjab, India

Relevant Coursework: Data Structures & Algorithms, Web Technologies, Database Management Systems, Software Engineering, Computer Networks,